Short-circuit, Protection Coordination & Arc-flash Study

30 MW IT hyperscale campus, Dalarna (SE3), Sweden · fictional · conceptual study

Item	Value
Document	DWU-DC-E-CALC-003, Rev P1a (concept; errata on sheet 6) — Python package `scprot` v0.1.0 (own IEC 60909 solver + pandapower cross-check)
Basis	DWU-DC-E-DB-001 Rev P1 §7–8, SLD-001…003 (both superseded by Rev P2); DWU-DC-E-CALC-002 (dcsizing) incl. rating changes B1–B11
Author / date	Donglin (Franklin) Wu — portfolio project 04 · 2026-10-25
Status	Concept — **not for construction**. Public standards, generic device curves, values [ASSUMED]; no employer/vendor data

1. Scope and method

IEC 60909-0:2016 equivalent-voltage-source method on a nodal model 130 kV grid → T1/T2 63 MVA → MSG-20 → 0.6 km hall feeders → HV-DH1 → 2.5 MVA block transformers → MSB → SBS/UOSB (bypass) → 800 A busway → rack PDU; genset 2 750 kVA on MSB-A. K_T/K_G, c_{max} 1.10/1.05, c_{min} 1.00/0.95, conductor θ_e 80 °C for $I_{k\,min}$; i_p method C, I_b with μ, I_{th} with m+n (T_k 1 s). Cross-check: pandapower 3.5 `calc_sc` — max case identical (Δ 0.00 %), min case 2–4 % lower (pandapower omits K_T for $I_{k\,min}$; lower value used for sensitivity). With MV cables/busduct omitted the solver reproduces the CALC-002 hand values exactly (53.8 kA LV, 13.0 kA 20 kV). Protection: IEC 60255-151 relays; **generic** IEC 60947-2-style LSI and IEC 60898-1 C-curve bands. Arc flash: IEEE 1584-2018 (arcflash-calc, validated against Annex D.1/D.2 in pytest); 20 kV outside the 15 kV model range → Lee method.

2. Findings

- **Normal (S0)**: MSG-20 13.0 kA, MSB-A 52.3 kA / i_p 122 kA — within 25 kA and 65 kA/143 kA. N-1 cases (S2 T2 out, S5 block/feeder out) give the same levels.
- **T1//T2 (S1) stays interlocked**: 22.2 kA is inside 25 kA but i_p 58.5 kA = 94 % of 62.5 kA and the asymmetrical breaking duty (τ ≈ 61 ms > 45 ms) is ≈ 95 % of rating; at S''_{kQ} 4 000 MVA i_p = 63.2 kA **exceeds** the making rating. Also: R-TX CT saturation, loss of section selectivity, OLTC circulating current. The hall-board coupler closed with both feeders (S5b) parallels T1/T2 too (20.9 kA) → interlock.
- **LV closed-transition retransfer (S3) not permitted**: grid + genset on MSB-A = 81.8 kA / i_p 187 kA > 65 / 143 kA. Keep open transition.
- **Busway**: 37.1 kA at the busway start > I_{cw} 36 kA → current-limiting MCCB with tested conditional rating (confirms B3). Tap-off 63 A devices see 25.1 kA, rack inlet 14.0 kA.
- **Genset island (S4)**: 27.4 kA initial, ≈ 11.9 kA sustained → genset ACB I_{sd} 2.5×I_r, SBS ACB I_{sd} 7.25 kA; all devices still trip. UPS on inverter (≈ 5.8 kA): rack MCB trips magnetically.
- **Arc flash**: MSB line side (cleared by the 20 kV relay) ≈ 96 cal/cm² → optical arc relay with MV intertrip cuts it to ≈ 3 cal/cm²; MSB bus 16.8 → 2.4–4.8 cal/cm² with maintenance mode / ZSI.
- **MV earthing**: compensated earthing keeps I_F ≈ 10.8 A (EPR ≈ 1 V) and detects R_F up to ≈ 10 kΩ; the 130 kV earth fault governs the earthing design.

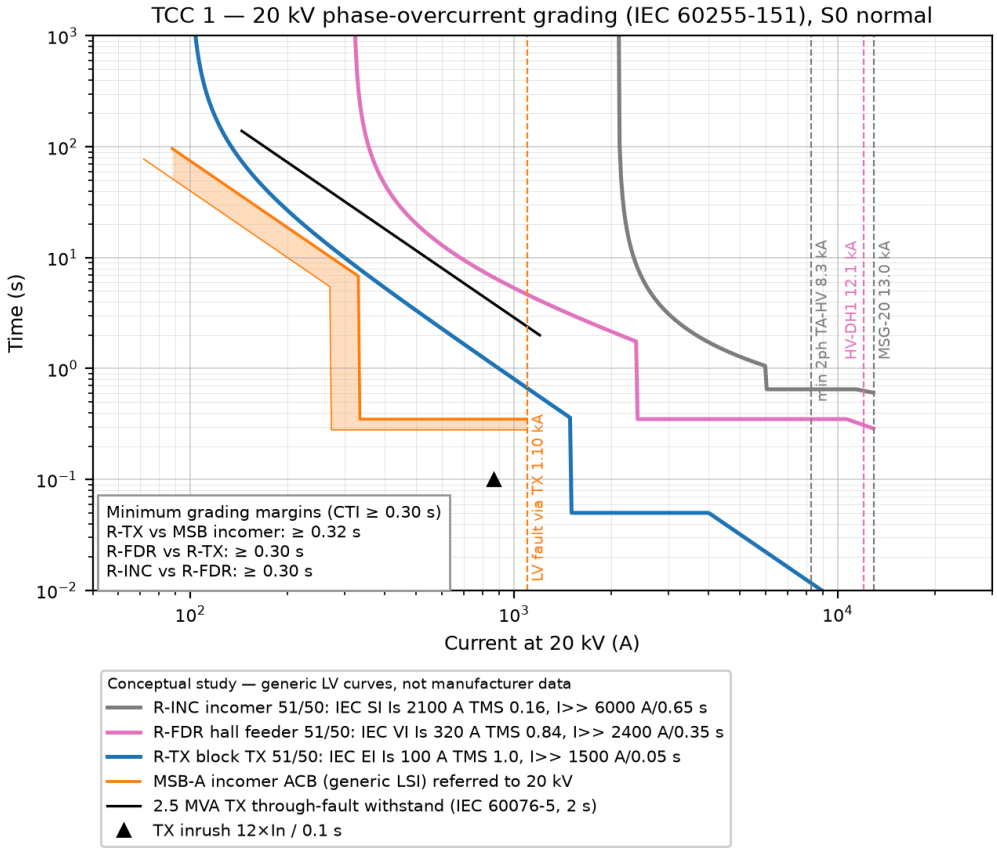
3. Short-circuit results (IEC 60909-0, kA)

Bus	S0 I''k max	i_p	I_b	I_{th} 1 s	κ	S0 I''k min	S1 T1//T2	S3 closed trans.	S4 genset	S5b hall cpl.
130 kV bus	13.3	32.9	13.3	13.5	1.75	6.7	13.3	13.4	—	13.3
MSG-20 sect. A (20 kV)	13.0	34.4	13.0	13.4	1.88	10.4	22.2	22.7	—	20.9
HV-DH1 sect. 1 (20 kV)	12.1	29.3	12.1	12.3	1.72	9.7	19.7	20.1	—	20.9
T-A HV terminals (20 kV)	11.8	27.2	11.8	11.9	1.63	9.5	18.9	19.3	—	20.1
MSB-A (400 V)	52.3	121.5	52.3	52.9	1.64	46.7	54.4	81.8	27.4	54.6
UOSB-A on bypass (400 V)	49.3	105.9	49.3	49.6	1.52	43.8	51.1	74.3	26.5	51.3
Busway A1 start (400 V)	37.1	67.9	37.1	37.3	1.29	32.4	38.1	49.1	22.6	38.2
Busway A1 end, 60 m (400 V)	25.1	39.2	25.1	25.1	1.11	20.7	25.5	29.2	17.8	25.5
Rack PDU inlet, 8 m 4G16 (400 V)	14.0	20.3	14.0	14.0	1.02	10.7	14.1	14.8	11.9	14.1

S0 = one 63 MVA transformer per 20 kV section, couplers open (also covers N-1 T2 out with coupler closed); min = S''_{kQ} 1 500 MVA, c_{min}, θ_e 80 °C.

UOSB/busway/rack values with UPS on static bypass (worst case). Full table incl. S2/S4min/S5: `results/short_circuit.csv`; pandapower comparison: `results/pandapower_crosscheck.csv`.

4. 20 kV protection coordination



Relay	CT	51: I_s A (×$I_{n,CT}$)	Curve / TMS	50: I>> A / t s	Grades with	Min. margin
R-INC 20 kV incomer (T1 LV)	2500/1	2100 (0.84)	IEC SI / 0.16	6000 / 0.65	R-FDR	0.30 s
R-FDR hall feeder (MSG-20)	400/1	320 (0.80)	IEC VI / 0.84	2400 / 0.35	R-TX	0.30 s
R-TX block transformer (HV-DH1)	150/1	100 (0.67)	IEC EI / 1.0	1500 / 0.05	MSB incomer ACB	0.32 s

TMS chosen as the smallest value giving ≥ 0.30 s over the full current range up to the maximum through-fault (S0). R-FDR I_s = 1.5 × N-1 feeder load (214 A); R-INC I_s = 1.2 × T1 rated current (1 732 A at 21 kV). R-TX: I>> 1 500 A above inrush (866 A) and the LV through-fault (1.10 kA); clears the 2-s withstand current (1 203 A) in 0.62 s. Incomer I>> 0.65 s can be reduced to ≈ 0.1 s by reverse blocking from feeder starts.

CT check (5P20, 1 A secondary, ALF' = ALF·(R_{ct}+$R_{b,n}$)/(R_{ct}+R_b), 2 × 20 m 2.5 mm² leads)

CT	Ratio	Class	R_{ct}+R_b Ω	ALF'	ALF' needed	I''k max kA	I_{th} CT kA	S0	if T1//T2
CT R-INC	2500/1	5P20 15 VA	8.33	55.2	5.2	13.0	250.0	✓	✓
CT R-FDR	400/1	5P20 15 VA	2.33	145.9	32.4	13.0	40.0	✓	✓
CT R-TX	150/1	5P20 10 VA	1.83	125.7	80.5	12.1	25.1	✓	✗

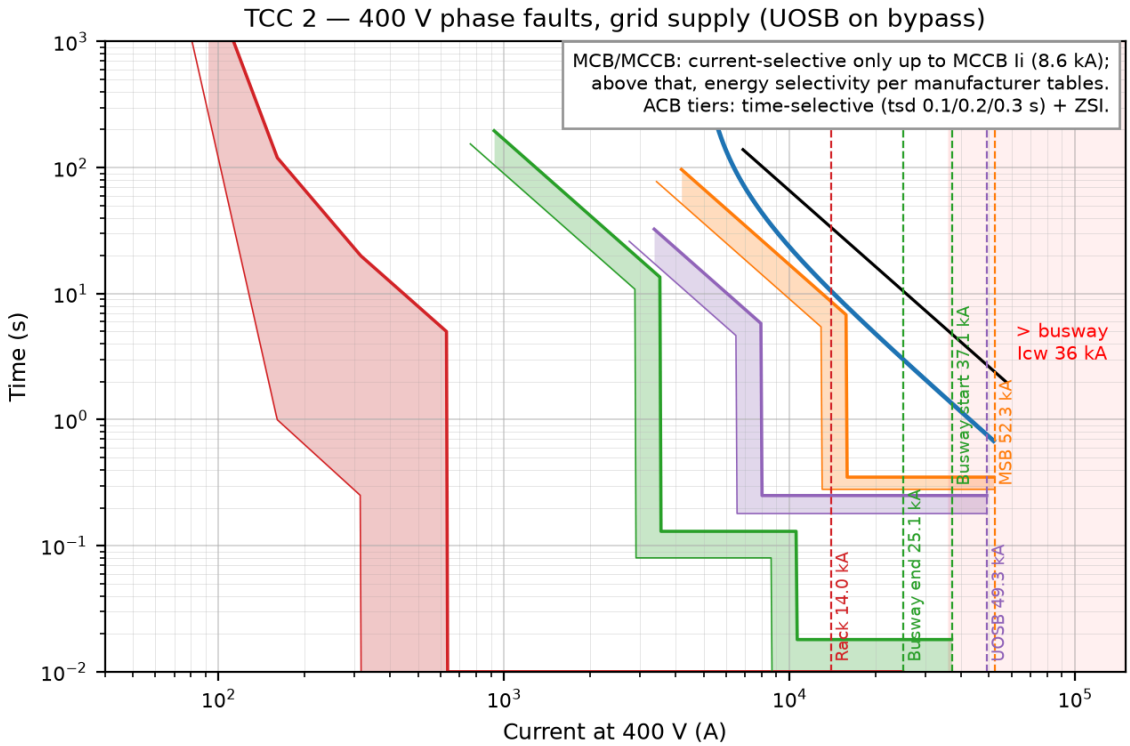
ALF' needed = max(I''k max/I_{pn}, 2·I>>/I_{pn}). R-TX 150/1 needs a thermal rating ≥ 25 kA 1 s (≈ 167 × I_{pn}) — specify; it would saturate if T1//T2 were allowed.

5. 20 kV neutral earthing — compensated vs low-resistance (conceptual)

	Compensated (Petersen coil + 10 A parallel R)	Low-resistance (NER 500 A, 10 s)
Earth-fault current	I_C 22.0 A per section (44.0 A coupled); residual $\|I_C-I_L\|$ + I_R ≈ **10.8 A**	≈ **500 A** (R_N 23.1 Ω via zig-zag)
Protection	67N admittance / transient (Y0>), feeders 1.0 s; 59N U0> backup 3 s	51N feeders 50 A / 0.3 s DT; incomer/NER 51N 50 A / 0.6 s
High-resistance fault detected	R_F ≲ 10 kΩ (U0 ≥ 10 %)	R_F ≲ 208 Ω (I0 ≥ 50 A)
EPR = r·I_F·R_E (r 0.6, R_E 0.2 Ω)	1.3 V ≪ 2·U_{Tp}(10 s) = 170 V	60 V < 2·U_{Tp}(0.5 s) = 440 V
Assessment	Arc self-extinguishes, low EPR and cable-screen stress; Swedish practice. Coil ≥ 50 A adjustable (both sections coupled ≈ 48 A).	Simple, fast and selective, but 500 A arcs damage cables/switchgear and every fault trips a feeder.

Touch voltage screened with the EN 50522 simplified criterion EPR ≤ 2·U_{Tp} (U_{Tp} per EN 50522 Fig. 4: 220 V at 0.5 s, 85 V at ≥ 10 s). The 130 kV earth fault governs (I_{k1} 8 kA [ASSUMED], r 0.5): EPR ≈ 800 V — OK for 0.1 s main protection (2·U_{Tp} 1 308 V) but not for 0.5 s back-up (440 V) → detailed earthing design with DSO data (SS-EN 61936-1, EN 50522, ELSÄK-FS 2022:1).

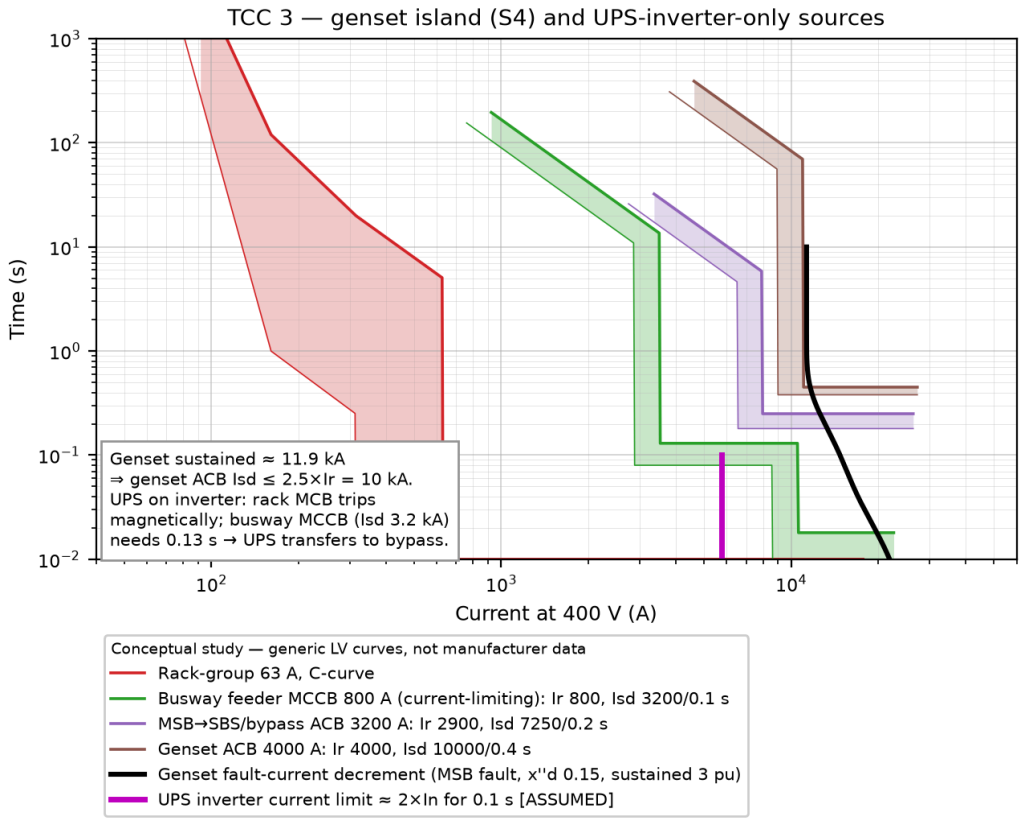
6. 400 V protection coordination



Device (generic trip unit)	I_r A	t_r @6I_r	I_{sd} A / t_{sd} s	I_i A	I_{cu}/I_{cw} kA	Notes
MSB-A incomer ACB 4000 A	3 600	3 s	14 400 / 0.3	OFF	65/65	ZSI to downstream ACBs; maintenance (ARMS) switch
MSB→SBS/bypass ACB 3200 A	2 900	1 s	7 250 / 0.2	OFF	65/65	I_{sd} ≤ genset sustained current
Busway feeder MCCB 800 A (current-limiting)	800	6 s	3 200 / 0.1	9 600	≥ 50 (I_{cu})	current-limiting; tested with busway
Genset ACB 4000 A	4 000	12 s	10 000 / 0.4	OFF	65/65	plus 51V / 32 / 40 generator protection
Rack-group / tap-off 63 A	63	C-curve	magnetic 5–10 I_n	—	≥ 25 (see §8)	IEC 60898-1 band

Downstream	Upstream	I''k max kA	Selectivity (grid supply, S0)
Rack-group 63 A	Busway feeder MCCB 800 A	25.1	current-selective to 8.6 kA; above → manufacturer energy-selectivity tables
Busway feeder MCCB 800 A	MSB→SBS/bypass ACB 3200 A	37.1	time-selective, min. band gap 0.05 s
MSB→SBS/bypass ACB 3200 A	MSB-A incomer ACB 4000 A	49.3	time-selective, min. band gap 0.03 s

7. Source-limited cases — genset island (S4 min) and UPS inverter



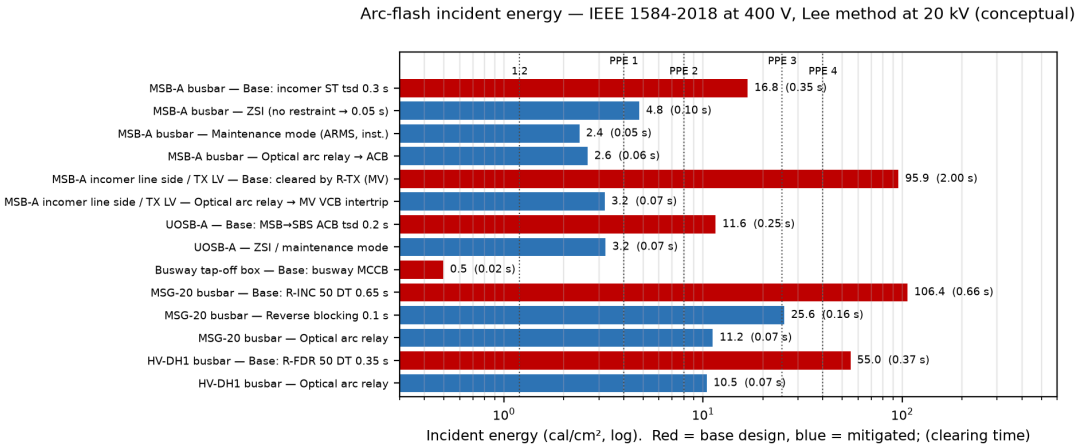
Device	Fault at	I''k initial kA	I at t_{sd} kA	Pick-up (+10 %) kA	Trips
Genset ACB 4000 A	MSB-A (400 V)	24.7	11.6	11.0	✓
MSB→SBS/bypass ACB 3200 A	UOSB-A on bypass (400 V)	23.9	12.1	8.0	✓
Busway feeder MCCB 800 A	Busway A1 end, 60 m (400 V)	15.3	13.8	10.6	✓
Rack-group 63 A	Rack PDU inlet, 8 m 4G16 (400 V)	9.4	6.1	0.6	✓

Genset decrement: x''_d 0.15, x'_d 0.22, T''_d 15 ms, T'_d 150 ms, sustained 300 % (PMG/AREP support) [ASSUMED]. UPS inverter ≈ 2 × I_n for 100 ms [ASSUMED]: rack MCB clears in 10 ms; a busway fault needs 0.13 s via the 800 A MCCB → UPS transfers to bypass to clear it. Min. L-PE fault at rack PDU (TN-S, c_{min}, 80 °C): 5.7 kA grid / 5.4 kA genset ≫ 630 A (10 × I_n) → disconnection within 0.4 s (IEC 60364-4-41).

8. Arc flash (conceptual — IEEE 1584-2018 / Lee)

Location	Case	Method	I_{bf} kA	I_{arc} kA	t s	E cal/cm²	AFB m	Arc rating
MSB-A busbar	Base: incomer ST tsd 0.3 s	IEEE 1584	52.3	27.7	0.350	16.8	3.18	PPE 3 (≥ 25 cal/cm²)
MSB-A busbar	ZSI (no restraint → 0.05 s)	IEEE 1584	52.3	27.7	0.100	4.8	1.45	PPE 2 (≥ 8 cal/cm²)
MSB-A busbar	Maintenance mode (ARMS, inst.)	IEEE 1584	52.3	27.7	0.050	2.4	0.94	PPE 1 (≥ 4 cal/cm²)
MSB-A busbar	Optical arc relay → ACB	IEEE 1584	52.3	27.7	0.055	2.6	1.00	PPE 1 (≥ 4 cal/cm²)
MSB-A incomer line side / TX LV	Base: cleared by R-TX (MV)	IEEE 1584	52.3	27.7	2.000	95.9	9.45	> 40 cal/cm² DANGER
MSB-A incomer line side / TX LV	Optical arc relay → MV VCB intertrip	IEEE 1584	52.3	27.7	0.067	3.2	1.13	PPE 1 (≥ 4 cal/cm²)
UOSB-A	Base: MSB→SBS ACB tsd 0.2 s	IEEE 1584	49.3	26.9	0.250	11.6	2.52	PPE 3 (≥ 25 cal/cm²)
UOSB-A	ZSI / maintenance mode	IEEE 1584	49.3	26.9	0.070	3.2	1.14	PPE 1 (≥ 4 cal/cm²)
Busway tap-off box (end)	Base: busway MCCB	IEEE 1584	25.1	17.3	0.018	0.5	0.26	< 1.2 cal/cm² (no arc-rated PPE req.)
MSG-20 busbar (20 kV)	Base: R-INC 50 DT 0.65 s	Lee	13.0	13.0	0.664	106.4	8.59	> 40 cal/cm² DANGER
MSG-20 busbar (20 kV)	Reverse blocking 0.1 s	Lee	13.0	13.0	0.160	25.6	4.21	PPE 4 (≥ 40 cal/cm²)
MSG-20 busbar (20 kV)	Optical arc relay	Lee	13.0	13.0	0.070	11.2	2.79	PPE 3 (≥ 25 cal/cm²)
HV-DH1 busbar (20 kV)	Base: R-FDR 50 DT 0.35 s	Lee	12.1	12.1	0.369	55.0	6.18	> 40 cal/cm² DANGER
HV-DH1 busbar (20 kV)	Optical arc relay	Lee	12.1	12.1	0.070	10.5	2.69	PPE 3 (≥ 25 cal/cm²)

8. Arc flash (cont.) — base design vs mitigation



LV: VCB, gap 32 mm, D 610 mm, 508 mm cube (switchgear); 25 mm / 457 mm / shallow panel (tap-off). Worst of full / reduced (VarCF) arcing current, clearing time read from the device band at each. Arc duration capped at 2 s. MV: Lee method (open-air theoretical maximum, D 910 mm) — conservative; IAC-tested switchgear protects front-access operators with doors closed. Arc rating labels follow NFPA 70E categories (4/8/25/40 cal/cm²); IEC 61482-2 APC classes are box-test based and must be selected separately.

9. Conclusions and required design changes (→ TODO_final_fixes.md B12–B21, DB-001 Rev P2)

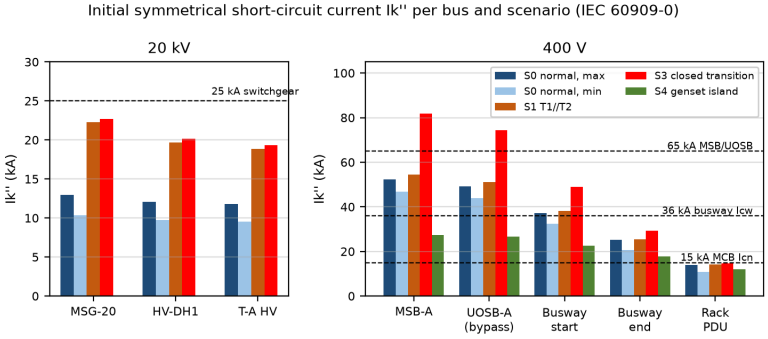
#	Item	Phase 1–2	Project 04	Reason
B12	MSG-20 T1//T2 and coupler make-before-break	interlocked (DB)	Confirmed: keep interlock; coupler auto-close only after incomer open (break-before-make)	S1 i_p 94 % of 62.5 kA; > 100 % at 4 000 MVA
B13	HV-DH1 hall coupler	N.O.	Interlock: close only with one hall feeder open	S5b parallels T1/T2 (20.9 kA)
B14	Genset retransfer	closed transition 'to be reconsidered'	**Open transition only**	S3 81.8 kA / 187 kA > 65 / 143 kA
B15	Busway feeder MCCB (B3)	Icc ≥ 55 kA	I_{cu} ≥ 50 kA, current-limiting, conditional busway rating ≥ 40 kA (tested combination)	37.1 kA at busway start > 36 kA
B16	Tap-off 63 A devices / rack PDU	MCB (n/a)	63 A MCCB I_{cu} ≥ 36 kA or tested cascade; PDU conditional SCCR ≥ 15 kA	25.1 kA busway end, 14.0 kA rack
B17	MSB/UOSB arc-flash mitigation	—	Optical arc relays + MV VCB intertrip; ZSI; maintenance (ARMS) switch	96 → 3.2 cal/cm² line side
B18	MSG-20 / HV-DH1	IAC AFLR 25 kA	Add optical arc protection + reverse-blocking busbar protection	Lee 106 → 11 cal/cm²
B19	LV trip-unit settings	LSIG (generic)	Table §6; genset ACB I_{sd} ≤ 2.5 I_r; genset sustained ≥ 300 % / 2 s in spec	genset island sensitivity
B20	20 kV CTs	—	R-INC 2 500/1, R-FDR 400/1, R-TX 150/1 5P20 with I_{th} ≥ 25 kA 1 s	CT check §4
B21	Petersen coil	per section	≥ 50 A adjustable + 10 A parallel resistor; 67N admittance	I_C 22 / 44 A

Ratings check. 9/14 pass; failures: MSB-A Icw 65 kA / 1 s (S3 LV closed transition), MSB-A Icm 143 kA (S3 LV closed transition), Busway 800 A Icw 36 kA / 1 s (S0 normal), Tap-off 63 A MCB Icn 15 kA (generic MCB) (S0 normal), Rack PDU inlet Icc 10 kA [ASSUMED typ.] (S0 normal). MSB 65 kA/143 kA and 20 kV 25 kA ratings stand (full model 52.3 kA vs 53.8 kA hand value; 13.0 kA confirmed).

Limitations. Generic LV curves (no manufacturer selectivity/cascade tables); assumed grid, cable, genset, UPS and earthing data; no motor/VFD/BESS fault contribution (BESS inverters ≈ 1.2 × I_n would add < 0.5 kA at 20 kV); IEEE 1584 not valid at 20 kV; touch voltage only screened (no soil model). Code:

`scripts/make_all.py` regenerates all figures/results; 23 pytest tests.

Fig. 1 — Initial short-circuit current per bus and scenario vs equipment ratings (§3)



PROJECT	TITLE	DOCUMENT NO.	REV	DATE	SHEET
D. Wu — Portfolio 30 MW IT DC campus, Dalarna SE3 (fictional)	**Short-circuit, protection & arc-flash study** Concept — not for construction	**DWU-DC-E-CALC-003**	**P1a**	2026-10-25	5 of 6 (A4)

Revision history and errata

Rev	Date	Description
P1	2026-10-24	First issue
P1a	2026-10-25	Errata only: cross-references to DB-001 Rev P2 / SLD Rev P2; superseded statements listed below. No calculation result changed.

Errata / superseded items (Rev P1a, 2026-10-25). Calculation results are unchanged. (1) "No BESS fault contribution" (limitations, §9) is **superseded by CALC-004**: with the 10 MW BESS (PCS limit 1.2 × I_n) MSG-20 I''_k is **13.3 kA**, worst case **23.1 kA (92 % of 25 kA)**; earth-fault I_c +≈1 A. (2) Changes B12–B21 are implemented in **DB-001 Rev P2** §7 and **SLD Rev P2** (relays and CTs on SLD-001 sh 1–2, LV trip settings and arc mitigation on SLD-003 sh 1–2).